********** MEMS 2024 – SOLUTIONS **********

Solution 1

a) With $\sigma_z = 0$ the first two Hooke's Law equations become:

$$E\varepsilon_x = \sigma_x - \nu\sigma_y$$
$$E\varepsilon_y = \sigma_y - \nu\sigma_x$$

Eliminating σ_y from these equations we get:

$$E\varepsilon_x = \sigma_x - \nu\left(E\varepsilon_y + \upsilon\sigma_x\right) \quad \Rightarrow \quad E\varepsilon_x + \upsilon E\varepsilon_y = \sigma_x(1-\upsilon^2)$$

and the RH equation is trivially rearranged to give the first of the required equations. The second equation is obtained by eliminating σ_x from the starting equations above.

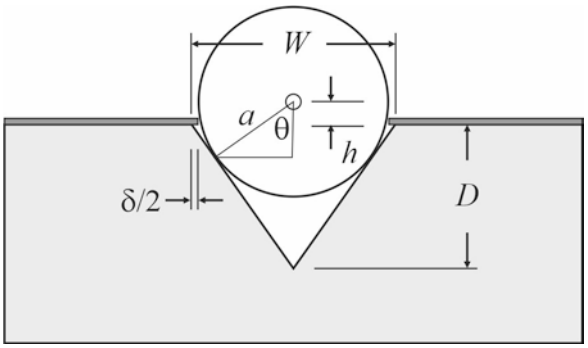
These equations are relevant to any plane stress problem, for example plate bending.

[6]

b) From the diagram below, the width of the final V-groove at the wafer surface needs to satisfy:

$$W/2 = a\sin\theta + (a\cos\theta - h)\cot\theta = (\sqrt{3}a - h)/\sqrt{2}$$

where we have used the fact that the angle between the V-groove sidewalls and the wafer surface is $\theta = \tan^{-1}\left(\sqrt{2}\right)$. With a = 62.5 μm and h = 5 μm, this gives W = 146.02 μm. The undercut is given by $\delta = \sqrt{6}D/S$ where S is the anisotropy, and the mask aperture needs to be narrower than the groove by this amount. With $D = \frac{W}{\sqrt{2}} = 103.25$ μm and S = 400, we get δ = 0.63 μm, so the required mask width is W_m = 145.39 μm.



[6]

c) Process steps for a surface micromachining process with 2 mechanical layers:

Deposit and pattern first sacrificial layer
Deposit and pattern first mechanical layer
Deposit and pattern second sacrificial layer
Deposit and pattern second mechanical layer

Remove sacrificial layer

Sacrificial layers: doped silicon dioxide
Mechanical layers: polysilicon
Depositions by LPCVD
Patterning by RIE
Sacrificial layer removal by wet chemical or vapour phase etching

[6]

d) For the case of on-axis, plane wave illumination, there is a sharp cut-off in the spatial frequency at NA/λ, beyond which the +1 and -1 diffraction orders do not pass through the project lens and no image can be formed. A single off-axis illumination beam also has a sharp cut-off, but in this case the limiting spatial frequency is higher, corresponding to a lower k_1 value.

In a practical system there are multiple illuminating beams at different angles, and in this case the modulation depth in the image falls off gradually beyond NA/λ so that a resist layer can be patterned provided the max and min intensities in the image are, respectively, higher than the resist sensitivity and lower than the resist threshold dose. The resolution limit in this case will depend on the resist contrast.

[5]

e) The bending equation for a cantilever with a transverse end load P is:

$$M = EI\frac{d^2v}{dx^2} = P(L-x)$$

where $v(x)$ is the deflection profile, x is distance along the cantilever measured from the root, L is the cantilever length, E is Young's modulus and I is the second moment of area. Assuming a rectangular cross-section, $I = wh^3/12$ where w and h are the cross-sectional dimensions, with h being in the direction of the load P.

Integrating twice, and applying the boundary conditions $v = 0$, $v' = 0$ at $x = 0$, the deflection profile is obtained as $v(x) = (PL^3/6EI)\cdot(3u^2 - u^3)$ where $u = x/L$.

Putting $u = 1$, the end deflection is $v_L = v(L) = PL^3/3EI$, so the stiffness is:

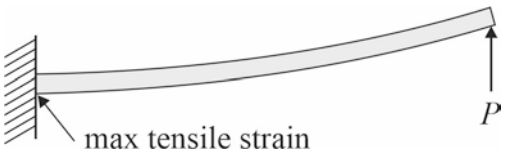
$$k = P/v(L) = 3EI/L^3$$

The axial strain varies as:

$$\varepsilon = -\frac{y}{R} \approx -y\frac{d^2v}{dx^2} = -\frac{yP(L-x)}{EI}$$

where y is the distance from the neutral axis in the direction of the applied load. The maximum tensile strain occurs when $y = -h/2$ and $x = 0$, and is given by $\varepsilon_{max} = PLh/2EI$. Combining this with the earlier result for the end deflection gives:

$$\varepsilon_{\max} = \frac{3v_L h}{2L^2}$$ [4]



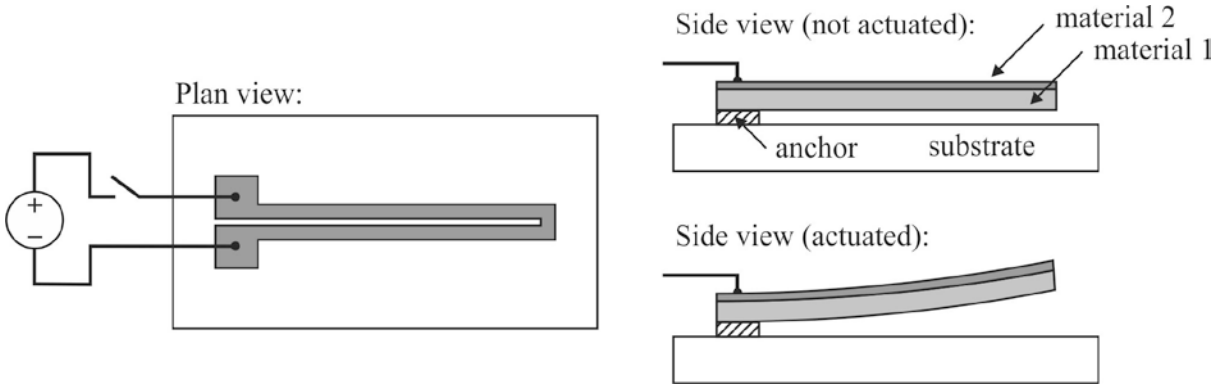
[2]

f) These equations are statements of Newton's second law ("f = ma") as applied to the gyroscope proof mass. The gyroscope is based on a two-axis resonator, and the equations are resolved along the principal axes which are uncoupled in the absence of any input rotation. The variables v_x and v_y are the proof displacements along x and y.

Considering either one of the equations: the first term is the inertial (i.e. "ma") term; the second term represents the damping force; the third term represents the restoring force due to the spring suspension; the fourth term represents cross-coupling between the axes which arises from the Coriolis force when there is an input rotation about the z-axis. It is this term that is key to the gyroscope action. The final term on the right is the actuator driving force.

Assuming the x-axis is driven at the resonant frequency and at constant amplitude A, the cross-coupling will cause excitation at the resonant frequency on the y-axis. In steady state the inertial and spring force terms in the y-axis equation will cancel out (resonance), and since $F_y = 0$ the y-axis equation will reduce to $c_y \dot{v}_y + 2m\Omega\dot{v}_x = 0$. From this it follows that the amplitude on the sensed axis is $B = -2m\Omega A / c$, and the ratio of amplitudes is $B / A = -2m\Omega / c$. [6]

g)



This kind of actuator relies on differential thermal expansion between the two materials in a bi-layer. Typically it comprises a beam with a thin film coating of a different material. When the structure is heated, differential thermal expansion leads to a bending moment, causing the structure to deflect upwards (if $\alpha_{beam} > \alpha_{film}$) or downwards ($\alpha_{beam} < \alpha_{film}$). Heating is generally effected by passing a current through either the beam or the thin film. A typical material system might be polyimide beam + metal film or silicon beam with metal film.

[5]

Solution 2:

a) We are told that the starting material is BSOI, so we can make the device using a standard BSOI process:

- Pattern the device layer by photolithography
- Transfer pattern through device layer by RIE
- Strip resist mask
- Release structure by wet/vapour-phase etching oxide layer

The perforations are needed so that the oxide etch can remove all of the oxide beneath the mass without completely undercutting the anchors (which need to remain fixed to the substrate via the oxide layer.

[6]

b) The suspension comprises four flexures in parallel so the overall stiffness will be 4× that of a single flexure, i.e. $k = 4 \times 12EI/l^3 = 4Ewh^3/l^3$. With E = 170 GPa, w = 10 μm, h = 2 μm, l = 600 μm we obtain $\boldsymbol{k}$ **= 0.252 N/m**.

The mass will be $m = \rho Ah$ where A is the proof mass area. With ρ = 2330 kg/m^3, A = 1.44 mm^2, h = 2 μm, this gives $\boldsymbol{m}$ **= 6.71 × 10^{-9} kg**. The resonant frequency is then obtained as:

$$\omega_0 = \sqrt{\frac{k}{m}} = \textbf{6,128 rad/s} \quad \textbf{or} \quad \textbf{975 Hz}$$

The static sensitivity (mass displacement per unit acceleration) is:

$$S_{mech} = \frac{1}{\omega_0^2} = \mathbf{26.6\ nm\ per\ m/s^2}$$

[10]

c) The proof mass, suspension beams and anchors will all contribute to the baseline capacitance. Each will contribute a capacitance of $C = \frac{A\varepsilon_r\varepsilon_0}{g}$ where A is the area and ε_r is the relative permittivity in the gap.

Proof mass: $A = 1.44 \times 10^{-6}\ m^2$; $\varepsilon_r = 1 \implies 6.372$ pF
Suspension beams: $A = 4 \times 10\ \mu m \times 600\ \mu m = 2.4 \times 10^{-8}\ m^2$; $\varepsilon_r = 1 \implies 106\ fF$
Anchors:

$$\begin{aligned} \varepsilon_r A &= 4[1 \times 100\ \mu m \times 100\ \mu m + (3.8-1) \times 80\ \mu m \times 80\ \mu m] \\ &= 1.12 \times 10^{-8}\ m^2 \implies 494\ fF \end{aligned}$$

where we have used g = 2 μm throughout. So, the total baseline capacitance is **6.97 pF**.

Under an applied acceleration a, the change in gap will be $\delta g = -\frac{a}{\omega_0^2} = -S_{mech}a$. So, for a = +10 m/s^2, δg = – nm and the shift in the proof mass capacitance is:

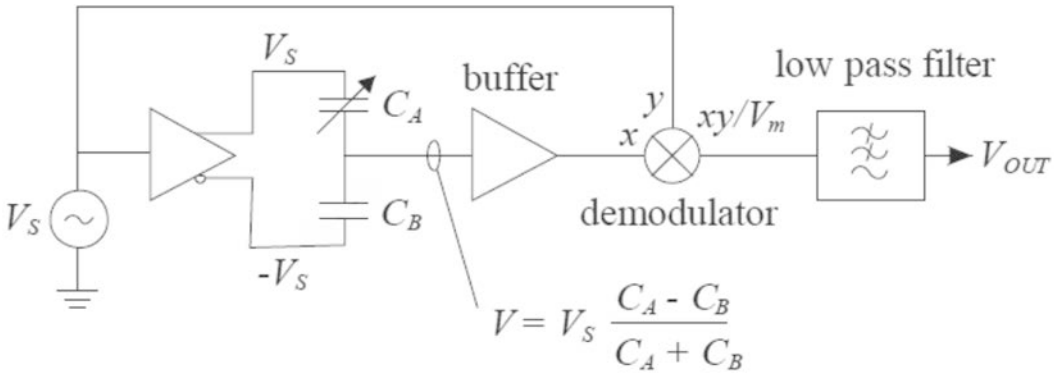
$$\delta C = \left(\frac{2}{2 - 0.266} - 1\right) \times 6.372 = +\mathbf{977\ fF}$$

Similarly, for a = –10 m/s^2:

$$\delta C = \left(\frac{2}{2 + 0.266} - 1\right) \times 6.372 = \mathbf{-748\ fF}$$

[8]

d) Any circuit that can make an AC capacitance measurement with sub-pF resolution is acceptable here. For example, the circuit shown in the lecture notes is a possible solution:



In this case C_A is the MEMS capacitance (to be measured) and C_B is a fixed reference capacitance equal to the baseline capacitance of the MEMS structure under zero applied acceleration. The final output voltage after the coherent demodulator and the low-pass filter will be:

$$V_{OUT} = \frac{V_0^2}{2V_m}\frac{C_A - C_B}{C_A + C_B}$$

[6]

Solution 3:

a) In a piezoelectric material, applied stress will generate dielectric polarization and hence surface charge. This is the direct piezoelectric effect. A given stress component may generate polarisation in more than one axis, and the general relationship between the stress and the polarisation is of the form:

$$P_i = d_{ij}\sigma_j$$

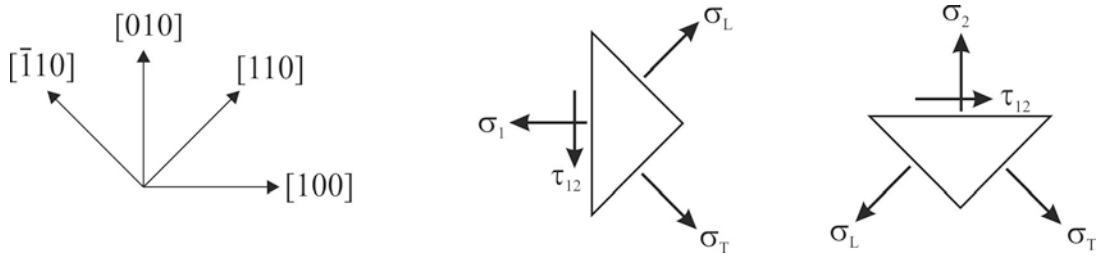
where d_{ij} is a 3 x 6 matrix of piezoelectric coefficients. (In the presence of electric field the above equation will also include a normal dielectric response term.) The effect is reversible, so an applied electric field will generate mechanical strain. The direct effect is used for transduction, while the inverse effect is used for actuation. The piezoelectric effect is observed only in non-centrosymmetric crystals, so silicon is not piezoelectric.

In a piezoresistive material, applied stress leads to a change in electrical resistivity. In general both the resistivity and the piezoresistive response will be anisotropic, and the generalised relation between electric field and current density is of the form:

$$\boldsymbol{E} = \left(\boldsymbol{\rho}_0 + \boldsymbol{\Pi}\cdot\boldsymbol{\sigma}\right)\cdot\boldsymbol{J}$$

where $\boldsymbol{\rho}_0$, $\boldsymbol{\Pi}$ and $\boldsymbol{\sigma}$ are all tensors. Usually symmetries allow these equations to be vastly simplified. [6]

b) Suggested approach is to consider the equilibrium of small prismatic elements (alternative would be to use coordinate rotation of stress tensor):



Resolving forces along [100] in the middle diagram (bearing in mind the relative lengths of the different sides) gives:

$$\sigma_1 \cdot \sqrt{2}a = \sigma_L a \cdot \frac{1}{\sqrt{2}} + \sigma_T a \cdot \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \sigma_1 = \frac{\sigma_L + \sigma_T}{2}$$

Similarly:

$$\sigma_2 = \frac{\sigma_L + \sigma_T}{2} \quad \text{and} \quad \tau_{12} = \frac{\sigma_L - \sigma_T}{2}$$ [6]

To obtain the next result, we need to apply the standard piezoresistive equations (see *Info for Candidates*) to the case where the electric field and current both lie along the [110] direction. In this case the ***E*** and ***J*** vectors referred to the usual <100> axes are $\boldsymbol{E} = (E/\sqrt{2}, E\sqrt{2}, 0)$

and $\boldsymbol{J} = (J/\sqrt{2}, J\sqrt{2}, 0)$, and we can write either the first or second piezoelectric equation as:

$$E/\rho_e = J[1 + \pi_{11}\sigma_1 + \pi_{12}\sigma_2] + J\pi_{44}\tau_{12}$$

Substituting for σ_1, σ_2 and τ_{12} from above, the resistivity becomes:

$$\frac{E}{J} = \rho_e\left\{1 + \pi_{11}\frac{\sigma_L + \sigma_T}{2} + \pi_{12}\frac{\sigma_L + \sigma_T}{2} + \pi_{44}\frac{\sigma_L - \sigma_T}{2}\right\} = \rho_e\{1 + \sigma_L\pi_L + \sigma_T\pi_T\}$$

The fractional change in resistivity, and hence in the overall resistance of the piezoresistor, is then:

$$\frac{\Delta R}{R} = \frac{\Delta\rho}{\rho_e} = \sigma_L\pi_L + \sigma_T\pi_T \quad \text{as required} \qquad [6]$$

c) Using the given equation, the maximum stress for the membrane in question at a unit applied pressure of 1 Pa will be $\sigma_{max} \approx 0.3 \times (2000/10)^2 = 12$ kPa. Also, from the given values for π_{ij}, the longitudinal and transverse piezoresistive coefficients are $\pi_L = 71.8 \times 10^{-11}$ Pa^{-1}, and $\pi_T = -66.3 \times 10^{-11}$ Pa^{-1}. [2]

One pair of piezoresistors (R_1, R_3 say) will experience predominantly longitudinal stress and will have a fractional resistance change of:

$$\Delta R_1/R_1 = \Delta R_3/R_3 = \sigma_{max}(\pi_L + \nu\pi_T) = +8.107 \times 10^{-6} = \alpha$$

The other pair will experience mainly transverse stress and will have a fractional resistance change of:

$$\Delta R_2/R_2 = \Delta R_4/R_4 = \sigma_{max}(\pi_T + \nu\pi_L) = -7.405 \times 10^{-6} = \beta \qquad [4]$$

The bridge output voltage will then be:

$$V_{out} = \frac{(R_1R_3 - R_2R_4)}{(R_1+R_2)(R_3+R_4)}V_S = \frac{(1+\alpha)^2 - (1+\beta)^2}{(2+\alpha+\beta)^2}V_S = \frac{\alpha-\beta}{2+\alpha+\beta}V_S$$

where we have used the fact that all four resistors have the same nominal value. With the above values for α and β, and with a bridge supply of $V_S = 5$ V, this gives $\boldsymbol{V_{out}}$ **= 39 μV/Pa**.

[6]

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Solution 4

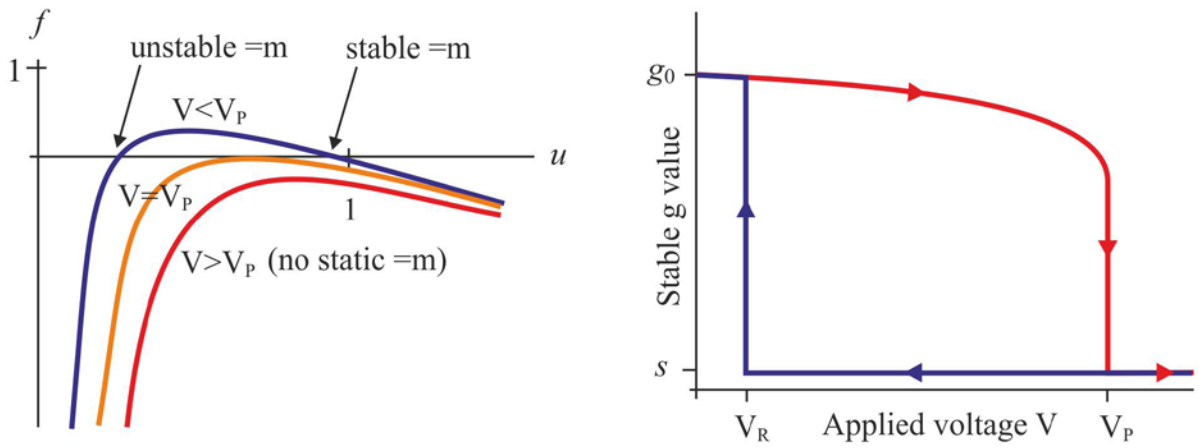
a) i) The electrostatic force on the moveable plate is $F_e = -A\varepsilon_0 E^2/2$ where $E = V/g$ is the electric field in the gap (this can be quoted). The only other force acting on the moveable plate is the spring force $F_k = k(g_0 - g)$, so the total force is:

$$F = F_e + F_k = k(g_0 - g) - \frac{A\varepsilon_0 V^2}{2g^2}$$

Dividing through by kg_0, and using the substitutions $a = A\varepsilon_0 V^2/2kg_0^3$ and $u = g/g_0$, the above becomes:

$$\frac{F}{kg_0} = f = 1 - u - \frac{a}{u^2}$$ as required. (1) [6]

a) ii)



The LH graph shows the variation of force with gap for different values of V (and hence a). When $V < V_P$ there are two equilibrium points (i.e. points where $f = 0$). One is stable, and the other is unstable, but the moveable plate naturally settles at the stable one (and never reaches the unstable one) if the applied voltage is increased from zero. For $V > V_P$ there is no equilibrium point, and the total force is always negative so the plate snaps down. Once this has occurred, the voltage must be reduced to the point where the electrostatic force at $g = s$ falls below the spring force. This results in hysteretic behaviour as shown in RH graph. [8]

b) The conditions at the point of snap-down are $f = 0$ and $\partial f/\partial u = 0$. Differentiating the force equation gives:

$$\frac{\partial f}{\partial u} = -1 + \frac{2a}{u^3}$$

So, when $\partial f/\partial u = 0$ we have $a = u^3/2$. Substituting this value of a into equation (1), and imposing the condition $f = 0$, we find that $u = 2/3$ at the point of snap-down. It follows that $a = 4/27$ at the point of snap-down, and so the snap-down voltage is given by:

$$\frac{4}{27} = \frac{A\varepsilon_0 V_P^2}{2kg_0^3} \quad \Rightarrow \quad V_P = \sqrt{\frac{8kg_0^3}{27A\varepsilon_0}} \qquad [4]$$

c) When the device is placed in series with the capacitor *C*, only a fraction of the drive voltage appears across the actuator, as determined by the potential divider rule. The electrostatic force becomes:

$$F_e = -\frac{A\varepsilon_0}{2g^2}\frac{V^2C^2}{(C + C_0/u)^2} = -\frac{A\varepsilon_0 V^2}{2(u+b)^2 g_0^2}$$

where we have used the fact that the actuator capacitance at an arbitrary position is C_0/u.

The total force is obtained as F = $F_e + F_k$, with F_k as in part a), and dividing through by kg_0 gives:

$$f = \frac{F}{kg_0} = 1 - u - \frac{a}{(u+b)^2} \quad \text{as required.} \qquad [4]$$

The '*b*' in the denominator of the last term reduces the electrostatic force. This raises curve *f*(*u*) and shifts the stationary point to the left.

We look for a value of *b* that will place the snap-down point at *g* = *s*, i.e. when the moveable plate is at its lowest possible position. If such a value of *b* exists, and is applied, then snap-down will never actually occur (since for lower applied voltages the force curve will always be higher and will have a stable equilibrium point, and by the time the snap-down voltage is reached the moveable plate will already be touching the spacers).

Applying the conditions *f* = 0 and $\partial f/\partial u = 0$ gives:

$$1 - u - \frac{a}{(u+b)^2} = 0 \quad \text{and} \quad -1 + \frac{2a}{(u+b)^3} = 0 \qquad (2)$$

Combining these relations we obtain the condition:

$$1 - 3u/2 - b/2 = 0$$

For this to apply at *g* = *s*, we require that $b = (2 - 3s/g_0)$. Larger values of b will also give stable behaviour (since increasing b always raises the force curve). The condition for stable behaviour is therefore:

$$b = \frac{C_0}{C} \geq (2 - 3s/g_0) \quad \text{or} \quad C \leq \frac{C_0}{(2 - 3s/g_0)} \qquad [4]$$

d) To find the pull-down voltage in the limiting case, we set $b = (2 - 3s/g_0)$ and $u = s/g_0$ in either of the equations (2) to find the value of *a*. This gives $a = 4(1 - s/g_0)^3$. The corresponding voltage is:

$$V = \sqrt{\frac{8k(g_0 - s)^3}{A\varepsilon_0}} \approx 3\sqrt{3}V_P \quad \text{if } s << g_0 \qquad [4]$$

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